

... [9–11], dietary habits [12], and the presence of pathological conditions [13,14]. Therefore, understanding the nonhomogeneous biomechanical behavior of these tissues is of paramount importance, particularly in a non-destructive manner. Such knowledge can inform medical diagnostics and treatments, potentially leading to better patient outcomes.

Considerable effort has been devoted into research aimed at identifying the nonhomogeneous biomechanical properties of biological tissues. Studies [15–18] have demonstrated the feasibility of estimating the mechanical properties of each layer in multi-layered biological tissues, provided that the thickness of each layer is known beforehand. However, in practice, determining the thickness of each layer prior to addressing the identification problem can be a challenging task. Furthermore, the interface between neighboring layers is seldom flat, a characteristic particularly pronounced in biological tissues. As such, a comprehensive solution requires knowledge of both the topology of the interface between the layers and the biomechanical properties of each layer. Additionally, the detection of tumors embedded within normal tissues introduces another layer of complexity. To accurately diagnose and plan treatment for these conditions, it is crucial to determine not only the elastic modulus values of these tumors but also their topology [19].

A general strategy for tackling this challenge involves identifying the spatial variation of material properties, a process that necessitates solving a large-scale optimization problem to optimize the elastic properties of every pixel in the domain of interest, which can be referred to as the implicit inverse method [20–23]. The implicit inverse method has been successfully employed in tumor detection using real clinical data [8,24]. However, this implicit inverse method has its limitations. It not only demands significant computational resources but is also prone to severe uniqueness issues, which can hamper the reliability of the results. Given that biological tissues are often known a priori to exhibit either layered structures or inclusion-embedded structures, it's feasible to explicitly and simultaneously identify the topology of each nonhomogeneous region and its associated material properties. This approach bypasses some of the computational hurdles of the implicit method. Another inherent drawback of the implicit inverse method lies in the quality of the reconstruction results, which can be compromised if the full-field displacement field cannot be measured with high accuracy. Studies [25,26] that employed the implicit inverse method to reconstruct the nonhomogeneous elastic property distribution of solids using surface deformation have demonstrated this issue. While these methods are capable of visualizing the nonhomogeneous region, the quality of the reconstruction is highly sensitive to noise. This sensitivity arises from a disparity between the large number of unknown optimization variables in the implicit inverse method and the sparsity of the deformation information on the surface. In contrast, the explicit inverse method significantly reduces the total number of optimization variables [27–29], potentially improving the quality of the reconstructed results. This strategy holds promise for enhancing the accuracy of nonhomogeneous biomechanical property identification in biological tissues.

This paper will present an explicit inverse method that utilizes surface deformation measurements to identify the mechanical behavior of two common nonhomogeneous structures: layered structures and inclusion embedded structures. The organization of this work is as follows: In the **Methods** Section, we provide a detailed explanation of the mathematical background underlying the explicit inverse approach. The **Results** Section presents a comparative study between the explicit and implicit inverse approaches, using various numerical examples. We discuss the obtained results in the **Discussion** Section and conclude the paper with a summary in the **Conclusion** Section.

Methods

Forward problem

Given the low elastic modulus of soft tissues, this study will consider finite deformation in the analysis. The strong form of the nonlinear elastic problem is stated as follows:

$$\begin{cases} \text{Div}(\mathbf{P}) = \mathbf{0} & \text{in } \Omega \\ \mathbf{u} = \mathbf{g} & \text{on } \Gamma_g \\ \mathbf{P} \cdot \mathbf{N} = \mathbf{T} & \text{on } \Gamma_T \end{cases} \quad (1)$$

where $\mathbf{P} = \mathbf{F}\mathbf{S}$ is the first Piola–Kirchhoff stress tensor where $\mathbf{F}$ is the deformation gradient and $\mathbf{S}$ is the second Piola–Kirchhoff stress tensor. Ω is the problem domain at the reference configuration. $\mathbf{g}$ is the prescribed displacement on the displacement boundary Γ_g . $\mathbf{T}$ is the prescribed traction on the traction boundary Γ_T . The following restrictions apply to Γ_g and Γ_T : $\Gamma_g \cup \Gamma_T = \partial\Omega$ and $\Gamma_g \cap \Gamma_T = \emptyset$. We assume the soft tissue of interest obeys the incompressible and neo-Hookean law:

$$\mathbf{S} = \mu(\mathbf{I} - \mathbf{C}^{-1}) + p\mathbf{C}^{-1} \quad (2)$$

where $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ is the Cauchy–Green tensor. p is the hydrostatic pressure.

The weak form of the nonlinear elastic forward problem is stated as: Find $[\mathbf{u}^h, p^h] \in \mathcal{M}^h \times \mathcal{P}^h$ such that

$$\mathcal{A}(\mathbf{w}^h, \mathbf{u}^h, q^h, p^h) = \int_{\Gamma_T} \mathbf{w}^h \cdot \mathbf{T} \, d\Gamma, \quad \forall [\mathbf{w}^h, q^h] \in \ell^h \times \mathcal{P}^h \quad (3)$$

$$\mathcal{A}(\mathbf{w}^h, \mathbf{u}^h, q^h, p^h) \equiv \int_{\Omega} \nabla \mathbf{w} : \mathbf{P} \, d\Omega + \int_{\Omega} q(J-1) \, d\Omega - \sum_{i=1}^{NE} \left(\tau \, \text{div}(\mathbf{F}^h \mathbf{S}^h), (\mathbf{F}^h)^{-T} \nabla q^h \right)_{\Omega_i} \quad (4)$$